

REWARD POISONING ATTACKS ON OFFLINE MULTI-AGENT REINFORCEMENT LEARNING

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Markov Game

- A finite-horizon n -player Markov game $G = (\mathcal{S}, \mathcal{A}, P, R, H, \mu)$ is:
 - $\mathcal{S}$ is the finite state space,
 - $\mathcal{A}$ is the finite joint action space,
 - $P_h : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ for $h \in [H - 1]$ is the transition kernel,
 - $\mathbf{R}_h : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^n$ for $h \in [H]$ is the mean reward function,
 - H is the finite horizon,
 - μ is the initial state distribution.

Offline Confidence Bound Learners

- The learners are uncertainty-aware. Given training set $\mathcal{D} = \left\{ \left(s_h^{(k)}, \mathbf{a}_h^{(k)}, \mathbf{r}_h^{0,(k)} \right)_{h=1}^H \right\}_{k=1}^K$ containing K episodes of length H , they estimate high confidence bounds on what Markov games (P and R) are plausible:
 - $\text{CI}_h^P(s, \mathbf{a}) = \left\{ P_h(s, \mathbf{a}) \in \Delta(\mathcal{A}) : \left\| P_h(s, \mathbf{a}) - \hat{P}_h(s, \mathbf{a}) \right\|_1 \leq \rho_h^P(s, \mathbf{a}) \right\}$ is the confidence set on transitions, where $\hat{P}_h(s'|s, \mathbf{a})$ is the maximum likelihood estimate of the transitions,
 - $\text{CI}_h^R(s, \mathbf{a}) = \left\{ R_{i,h}(s, \mathbf{a}) \in [-b, b] : \left| R_{i,h}(s, \mathbf{a}) - \hat{R}_{i,h}(s, \mathbf{a}) \right| \leq \rho_h^R(s, \mathbf{a}) \right\}$ is the confidence set on rewards, where $\hat{R}_{i,h}(s, \mathbf{a})$ is the maximum likelihood estimate of the rewards.
- The learners are rational in the sense they do not take dominated actions.
 - $\pi : Q_{i,h}^\pi(s, \pi_{i,h}(s), a_{-i}) \geq Q_{i,h}^\pi(s, (a_i, a_{-i})) + \epsilon$ for each $a_i \in \mathcal{A}_i, a_i \neq \pi_{i,h}(s)$ and $a_{-i} \in \mathcal{A}_{-i}$ is the policy a learner will use should such π exist. π satisfying such condition for all $i \in [n], h \in [H], s \in \mathcal{S}$ is called an ϵ -strict Markov perfect dominant strategy equilibrium (MPDSE).

Policy Poisoning: Threat Model

- Knowledge of the attacker:
 - The attacker has access to the original data $\mathcal{D}$,
 - The attacker knows the learners' ρ_h^P and ρ_h^R parameters, but not their exact learning algorithm.
- Available actions of the attacker:
 - The attacker is allowed to modify the rewards before the learner observe $\mathcal{D}$.
- Attacker's goals:
 - The attacker has a pre-specified target policy $\pi^\dagger$ that it wants the learners to learn.
 - The attacker wants to poison $\mathcal{D}$ to ensure $\pi^\dagger$ is the unique ϵ -strict MPDSE while minimizing its poisoning cost $C(r^0, r^\dagger)$.

Reward Poisoning Attack Formulation

- The attacker's problem is:

$$\begin{aligned} \min_{r^\dagger} C(r^0, r^\dagger) \\ s.t. R_{i,h}^\dagger(s, \mathbf{a}) &= \frac{1}{N_h(s, \mathbf{a})} \sum_{k=1}^K r_{i,h}^{\dagger,(k)} \mathbb{1}_{\{s_h^{(k)}=s, \mathbf{a}_h^{(k)}=\mathbf{a}\}}, \forall h, s, i, \mathbf{a} \\ \text{CI}_{i,h}^{R^\dagger}(s, \mathbf{a}) &= \left\{ R_{i,h}(s, \mathbf{a}) \in [-b, b] : \left| R_{i,h}(s, \mathbf{a}) - R_{i,h}^\dagger(s, \mathbf{a}) \right| \leq \rho_h^R(s, \mathbf{a}) \right\}, \forall h, s, i, \mathbf{a} \\ \underline{Q}_{i,h}(s, \mathbf{a}) &= \min_{R_{i,h} \in \text{CI}_{i,h}^{R^\dagger}(s, \mathbf{a})} R_{i,h} \\ &+ \begin{cases} 0 & \text{if } h = H \\ \min_{P_h \in \text{CI}_h^P(s, \mathbf{a})} \sum_{s' \in \mathcal{S}} P_h(s') \underline{Q}_{i,h+1}(s', \pi_{h+1}^\dagger(s')) & \text{if } h < H, \forall h, s, i, \mathbf{a} \end{cases} \\ \bar{Q}_{i,h}(s, \mathbf{a}) &= \max_{R_{i,h} \in \text{CI}_{i,h}^{R^\dagger}(s, \mathbf{a})} R_{i,h} \\ &+ \begin{cases} 0 & \text{if } h = H \\ \max_{P_h \in \text{CI}_h^P(s, \mathbf{a})} \sum_{s' \in \mathcal{S}} P_h(s') \bar{Q}_{i,h+1}(s', \pi_{h+1}^\dagger(s')) & \text{if } h < H, \forall h, s, i, \mathbf{a} \end{cases} \\ \underline{Q}_{i,h}(s, (\pi_{i,h}^\dagger(s), a_{-i})) &\geq \bar{Q}_{i,h}(s, (a_i, a_{-i})) + \epsilon, \forall h, s, i, a_{-i}, a_i \neq \pi_{i,h}^\dagger(s) \\ \mathbf{r}_h^{\dagger,(k)} &\in [-b, b]^n, \forall h, k \end{aligned}$$



Feasibility

- The attack is feasible if $\epsilon \leq 2b - (H + 1) \rho_h^R(s, \mathbf{a}), \forall h \in [H], s \in \mathcal{S}, \mathbf{a} \in \mathcal{A}$.
- When $C(r^0, r^\dagger) = \left\| r^0 - r^\dagger \right\|_1$, the problem can be formulated as a linear program.
- There exists a dataset $\mathcal{D}$ such that the attack cost is at least $H |\mathcal{S}| \underline{N} n |\mathcal{A}_1|^{n-1} (2b + 2\rho + \epsilon)$, where $\underline{N}$ is the minimum data coverage of any specific $(h, s, \mathbf{a})$ tuple, and assuming $|\mathcal{A}_i| = |\mathcal{A}_1| \forall i \in [n]$.
- To speed up the optimization, a Greedy Backward-Induction Attack can be used to compute a possibly sub-optimal feasible solution to the attacker's problem.

Example: Public Good Game

- Consider the iterated public goods game in which all participants need to volunteer in order to have a public good provided, for example, donations to environmental causes, private investment for startups, and voluntary vaccination to achieve herd immunity.
- As a simple two-period example, the reward matrices and transitions are given by:

$$\begin{array}{c|cc|cc} & \hat{R}_1(0, \cdot) & & \hat{R}_2(0, \cdot) & \\ \hline \mathcal{A}_1/\mathcal{A}_2 & V & N & V & N \\ \hline V & \mathbf{1, 1} & 0, 0 & \mathbf{1, 1} & 0, 0 \\ \hline N & 0, 0 & 0, 0 & 0, 0 & 0, 0 \end{array}$$

$$\begin{array}{c|cc|cc} & \hat{R}_2(1, \cdot) & & \hat{R}_2(2, \cdot) & \\ \hline \mathcal{A}_1/\mathcal{A}_2 & V & N & V & N \\ \hline V & \mathbf{0.5, 0.5} & -0.5, 0 & \mathbf{0, 0} & -1, 0 \\ \hline N & 0, -0.5 & 0, 0 & 0, -1 & 0, 0 \end{array}$$

$$\begin{aligned} \mathbb{P}\{s' = 0 | s, \mathbf{a} = (V, V)\} &= 1, \mathbb{P}\{s' = 1 | s, \mathbf{a} \in (V, N), (N, V)\} = 1, \\ \mathbb{P}\{s' = 2 | s, \mathbf{a} = (N, N)\} &= 1, \mu(s = 0) = 1. \end{aligned}$$

- The original game has a Pareto dominant Markov perfect Nash equilibrium in which all participants volunteer in all stages and the public good is always provided.
- Suppose the attacker wants to sow social distrust and prevent the public good, the attacker can produce fake training data to misinform the participants about the benefits and costs and as a result incentivize them to choose not to volunteer. $\pi^\dagger = (N, N)$ in all stages. The estimated reward matrices in the poisoned game are given by,

$$\begin{array}{c|cc|cc} & R_1^\dagger(0, \cdot) & & R_2^\dagger(0, \cdot) & \\ \hline \mathcal{A}_1/\mathcal{A}_2 & V & N & V & N \\ \hline V & 0.344, 0.344 & -0.005, 0.359 & 0.346, 0.346 & -0.005, 0.356 \\ \hline N & 0.359, -0.005 & \mathbf{0.005, 0.005} & 0.356, -0.005 & \mathbf{0.005, 0.005} \end{array}$$

$$\begin{array}{c|cc|cc} & R_2^\dagger(1, \cdot) & & R_2^\dagger(2, \cdot) & \\ \hline \mathcal{A}_1/\mathcal{A}_2 & V & N & V & N \\ \hline V & 0.213, 0.213 & -0.5, 0.223 & -0.005, -0.005 & -1, 0.005 \\ \hline N & 0.223, -0.5 & \mathbf{0, 0} & 0.005, -1 & \mathbf{0, 0} \end{array}$$

- In the poisoned game, action V is strictly dominated in all stages. When $K = 16, \epsilon = 0.01$ and $\rho = 0$, the attack cost is approximately 10.1.

Remarks

- We expose the danger of reward poisoning in offline multi-agent reinforcement learning (MARL), whereby an attacker can modify the reward vectors to different learners in an offline data set while incurring a poisoning cost.
- Based on the poisoned data set, all rational learners using some confidence-bound-based MARL algorithm will infer that a target policy – chosen by the attacker and not necessarily a solution concept originally – is the Markov perfect dominant strategy equilibrium for the underlying Markov Game, hence they will adopt this potentially damaging target policy in the future.
- We characterize the exact conditions under which the attacker can install a target policy. We further show how the attacker can formulate a linear program to minimize its poisoning cost.